Monitoring Overhaul

SolariNet Monitoring · why the graphs were blank, and a robust four-tier monitoring infrastructure · 2026-07-03

> **Headline.** The dashboard's graphs "never showed data" for two unrelated reasons — one on the read path (fixed) and one because an entire monitoring tier was never producing rows (now producing). Host metrics were always being stored; the fleet sparklines just weren't served them, and the reachability tier had no reporting vantage. Both are resolved, plus network-throughput history and a portable remote-outpost tier now live on xenon + benzene.

Deep-dive: why graphs were blank

A workflow of parallel readers traced the full path (collect → transmit → ingest → schema → API → charts). The leading "nothing is persisted" hypothesis was *refuted* for host metrics and *confirmed* for probes:

SYMPTOM	ROOT CAUSE	FIX
Primary dashboard fleet/ heat sparklines blank	`/api/nodes` returned no `hist`; every sparkline bound to `[]` (100k+ hostHistory rows existed, just weren't served)	List endpoint embeds a 40-sample cpu/ram/disk/net series per node (one windowed query)
Node-detail cpu/ ram/disk charts blank	The merged detail-fetch code (`screens.jsx`) was **stale in the web root** — never deployed	Deployed; detail page fetches per-metric history (network-first SW serves it fresh)
"Net" trend permanently empty	Empty by construction — agent reported iface rx/tx but hostHistory had no network column	New `hostHistory.netKbps` (migration 005) aggregated server-side, whitelisted + charted

Reachability / RTT / loss graphs empty	**Blocker:** probeHistory = 0 — 18 targets configured but no monitor vantage was reporting	Monitor outposts on xenon + benzene now report; 2 vantages × 18 targets

The four monitoring tiers

TIER	WHAT	STATE
On-host agent	`solariClient` → host metrics (cpu/ram/disk/net/procs/services) → hostCurrent + hostHistory	live; net history added
Orchestrator	`solariServer` ingest → MariaDB (append-only history), alerts, lease/failover	live; net aggregation added
Remote outposts	`solariMonitor` vantages → ICMP/TCP/UDP reachability (RTT/jitter/loss) → probeCurrent/probeHistory	live on xenon + benzene; portable
SNMP	poll managed network gear IF-MIB interface counters -> gearInterface{Current,History}	live (solariSnmpPoll timer); needs SNMP enabled on the device

Remote monitoring outposts

A monitor "outpost" is a portable prober that checks reachability from its vantage and reports to the server over mutual TLS. Multiple vantages give path-diverse reachability (a target down from one outpost but up from another localizes the fault).

```
# build a portable outpost bundle (monitor + matched TLS libnng) for an arch
deploy/build-outpost.sh x86_64        # or: arm64  (Pi Zero 2 W-class)

# deploy an outpost (local, or --host user@remote)
deploy/deploy-monitor.sh --name benzene-mon --host jason@benzene.akoria.net \
    --server tls+tcp://xenon:7701 --server-ip 10.0.0.20 --op admin
```

- **Enrollment:** each outpost gets a cert with CN `monitor.<name>` — the server derives role from the CN and *only* a monitor may send `MONITOR_REPORT` (a

`client.*` CN is rejected `ERR_AUTH_ROLE`). Signed via the bridge `SIGN`; the CA key never leaves the server.

- **Targets:** pulled from the `probeTarget` table into `monitor.conf` (`target = icmp:host` / `proto:host:port`). ICMP works unprivileged on Linux.
- **Portability:** the outpost carries its own TLS-enabled `libnng.so.1` and `LD_LIBRARY_PATH`s it — some distros (openSUSE) ship an nng whose TLS transport is ABI-incompatible, which is exactly why benzene initially spooled instead of reporting. Built on a debian:bookworm glibc floor so the same binary runs on Pi OS / appliances.
- **Mesh (available, not yet enabled):** the `[peer]` gossip + rendezvous-hash ownership lets N outposts split targets so they don't all probe everything — useful as the fleet of Pi outposts grows.

SNMP tier (managed gear)

The server polls each `networkGear`'s IF-MIB interface counters by **numeric OID** via the net-snmp CLI (`snmpbulkwalk`) — no MIB files and no `libnetsnmp` link, so the core stays dependency-free (gated by `SOLARI_WITH_SNMP`). In/out throughput rates are computed server-side from the octet deltas between samples and stored in `gearInterfaceCurrent` + `gearInterfaceHistory`.

- **Driver:** `solariSnmpPoll` run by `solarinet-snmp.timer` every 60s — deliberately OUT of the failover-critical server loop.
- **Per-gear community:** `networkGear.snmpCommunity` (falls back to the poller default).
- **Dashboard:** `/api/gear`, `/api/gear/{id}/interfaces`, `/api/gear/{id}/history` + a managed-gear throughput panel.
- **Verified** end-to-end against a local `snmpd` test target (real interface rates). Live data from your gear needs SNMP enabled on the device (e.g. the UniFi controller) and its community set.
- **Follow-up:** LLDP-over-SNMP neighbor discovery (the `topoSnmpFetchLldp` seam) is stubbed; the IF-MIB interface poller is the implemented feature.

Configuration surfaces

WHERE	KNOBS
client.conf	`[schedule] sampleIntervalSec`, watched processes/logs, spool db
monitor.conf (`deploy/monitor.conf.sample`)	`[probe] roundIntervalSec`, `target=` list, `[server] primaryUrl/failoverUrl`, `[peer]` mesh, spool db
server.conf	ingest/control/pub bind URLs, `[ca]` block
probeTarget table	the canonical reachability target set (host/port/proto/replFactor) — outposts render from it

Verification

- Fleet sparklines: `/api/nodes` returns 40-point cpu/ram/disk/net series per node.
- Net history: `netKbps` writes per report (xenon ~1320 Kbps) and serves via `/history?metric=netKbps`.
- Reachability: `probeCurrent` = 2 vantages × 18 targets; `probeHistory` accumulating; `/api/probes` returns per-target multi-vantage RTT/loss.
- Server unit tests: 23/23. PHP lint + JSX transpile clean.

Follow-ups (scoped, not yet built)

> These came out of the deep-dive and are the next increments toward "robust":

- **SNMP LLDP discovery** — the IF-MIB interface poller is built; the LLDP-over-SNMP neighbor walk (topology auto-discovery of managed gear) remains a stub to implement.
- **Ingest timestamp from frame header** — stamp hostHistory/probeHistory `sampledAt` from the frame's `sendTimeUnixMs` so store-and-forward spool replays land on the real sample time.

- **History self-maintenance** — systemd timers for `roll_history.sql` (monthly partition creation) + `retention.sql` (nightly prune); surface retention as a server.conf knob.
- **Read-path downsampling** — the `/history` endpoint truncates wide windows to the first N; bucket/downsample over the requested range instead. Give Sparkline an explicit empty/collecting state.
- **One-click target adoption** — wire `CTRL_ADOPT_TARGET` so dashboard-added probe targets start producing series without redeploying outposts.
- **Pi-Zero outpost fleet** — the arm64 bundle is built; deploy to the low-power Pis and enable the `[peer]` mesh so they split target ownership.
- **Infra self-monitoring** — alert on ingest lag, stalled series, spool depth, partition health, vantage quorum, so a silent write-side failure surfaces instead of manifesting as blank graphs.

SolariNet Monitoring Overhaul — 2026-07-03. Graphs across the primary dashboard and detail pages now present data; reachability outposts live on xenon + benzene and portable to Pi Zero 2 W-class hosts.